

# Categorical Grounding for the Shielding Calculators

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# 1 Why This Companion Exists

`physics-architecture.tex` gives the unified view. The hidden object is Ramsey magnetic shielding: the electronic current response to an applied magnetic field and to a nuclear magnetic moment [1, 2]. The calculators are shadows of that object. Many of those shadows are made from the same classical kernel,

$$D_{ab}(\boldsymbol{\rho}) = \frac{3\rho_a\rho_b - \delta_{ab}\rho^2}{\rho^5},$$

where  $\boldsymbol{\rho}$  is the vector from a source to the observed nucleus,  $\rho = |\boldsymbol{\rho}|$ , and  $\delta_{ab}$  is one when the Cartesian directions  $a$  and  $b$  are the same.  $D_{ab}$  is the second derivative of  $1/\rho$ . It is the same shape as a magnetic dipole field and as a point-charge electric-field gradient away from the charge.

The field usually reads the same physics another way. It reads by nucleus and by geometry:  $^{13}\text{C}^\alpha$  and  $^{13}\text{C}^\beta$  report Ramachandran state;  $^1\text{H}^\text{N}$  reports hydrogen bonding and electric field;  $^{15}\text{N}$  is multi-determinant and often harder; aromatic-near protons report ring currents; methyl and side-chain atoms report  $\chi$  rotamers and contacts. SHIFTX, SPARTA, CamShift, ProCS15, PPM, and UCBSHift all live in this categorical world [11, 9, 10, 13, 12, 14, 38, 39]. They are geometry-to-shift machines.

This document keeps that categorical view, but it ties each category back to first principles and to the actual calculators. The central discipline is the same as in `physics-architecture.tex`: except for LarsenHBond's DFT-derived lookup tables, the calculators emit geometric kernels, not chemical shifts and not physical shieldings in ppm. Calibration is the step that can attach a coefficient and test correlation against DFT or experiment.

## 2 The Observable, Seen Categorically

For a nucleus  $I$  in an applied field  $B_0$ , the field at the nucleus is

$$B_{\text{nuc}} = (\mathbf{I} - \boldsymbol{\sigma}_I)B_0.$$

$\boldsymbol{\sigma}_I$  is the shielding tensor at that nucleus. The isotropic shielding is

$$\sigma_{\text{iso},I} = \frac{1}{3} \text{tr } \boldsymbol{\sigma}_I.$$

The reported chemical shift is a referenced and signed version of this number, often written in a calibrated predictor as

$$\delta_I = b_I - a_I \sigma_{\text{iso},I}.$$

$\delta_I$  is the observed chemical shift for atom type  $I$ .  $a_I$  and  $b_I$  are fitted slope and intercept terms for that nucleus or atom class. The minus sign is the NMR convention: more shielding usually means a smaller chemical shift.

The code does not stop at the trace when a tensor exists. It stores a  $3 \times 3$  matrix as one scalar  $T_0$ , three antisymmetric components  $T_1$ , and five symmetric traceless components  $T_2$ . In categorical language,  $T_0$  is the isotropic shadow most shift predictors target.  $T_2$  is the orientation-dependent shadow: the part that knows whether a carbonyl group, aromatic plane, hydrogen bond, or electric-field gradient has turned.  $T_1$  is kept because several geometric kernels are asymmetric before the observable is reduced.

The old measures were invented because particular nuclei moved in ways a simpler picture could not explain. Ring current was named because aromatic hydrocarbon protons and protein protons near aromatic side chains were shifted by a delocalized  $\pi$ -electron response [19, 21, 22, 23]. Bond and group anisotropy were named because a covalent group is not magnetically spherical [20]. Electric-field shielding entered because polar groups and solvent changed shifts through the field at the nucleus and

through the field-gradient tensor [26, 27, 28]. Hydrogen-bond terms were isolated because amide protons and amide nitrogens move with hydrogen-bond length and orientation [32, 33, 14]. The calculators preserve those named measures as explicit geometric objects.

## 3 Backbone Carbons and the Ramachandran Surface

### 3.1 $^{13}\text{C}^\alpha$ , $^{13}\text{C}^\beta$ , and $^1\text{H}^\alpha$

The first categorical success of protein chemical shifts was not a global shielding derivation. It was an empirical observation: secondary structure leaves systematic chemical-shift offsets. Williamson’s secondary-structure survey and the chemical-shift index turned backbone chemical shifts into a practical structural readout [5, 6]. Spera and Bax then made the connection more geometric by correlating  $^{13}\text{C}^\alpha$  and  $^{13}\text{C}^\beta$  shifts with backbone conformation [7]. TALOS inverted the same fact: given chemical shifts, infer likely backbone  $\phi, \psi$  angles by database search [8]. SPARTA then inverted it back: given structure, predict backbone shifts by finding database triplets close in residue type and torsion space [9].

The first-principles reason this works is local but not scalar.  $C^\alpha$  is the tetrahedral junction between the amide nitrogen, carbonyl carbon, hydrogen, and side chain. Changing  $\phi$  rotates the previous carbonyl and amide environment around the  $N - C^\alpha$  bond. Changing  $\psi$  rotates the next carbonyl environment around the  $C^\alpha - C'$  bond. Those rotations move nearby anisotropic bonds, charges, hydrogen bonds, and solvent fields relative to the  $C^\alpha$  shielding tensor.  $C^\beta$  adds the side-chain axis, so it feels both  $\phi, \psi$  and  $\chi_1$ .  $H^\alpha$  sits on the same centre and is especially exposed to ring currents, hydrogen bonding, and local electric fields.

A first-order local expansion makes the geometry point explicit:

$$\Delta\sigma_I \approx \frac{\partial\sigma_I}{\partial\phi}\Delta\phi + \frac{\partial\sigma_I}{\partial\psi}\Delta\psi + \frac{\partial\sigma_I}{\partial\chi_1}\Delta\chi_1 + \Delta\sigma_{\text{env},I}.$$

$\Delta\sigma_I$  is the shielding change at nucleus  $I$ . The first three terms say how much shielding changes when the backbone or side-chain torsions move.  $\Delta\sigma_{\text{env},I}$  collects non-torsional environmental changes: ring currents, hydrogen bonds, electric fields, dispersion contacts, and solvent. The expansion is not a formula the code evaluates. It is the categorical map the field has used for decades.

The current shielding-kernel suite does not contain a Spera–Bax ( $\phi, \psi$ ) lookup surface. That absence matters. The calculators touch these nuclei through the mechanisms that  $\phi, \psi$  moves in Cartesian space. McConnell sums nearby covalent bond anisotropy, including peptide  $C = O$ , peptide  $C - N$ , side-chain  $C = O$ , aromatic, and other categories. APBSField and WaterField report the electric field and field-gradient environment. HBond and LarsenHBond report hydrogen-bond axes or DFT grid terms when the  $H \cdots O$  geometry is present. The aromatic calculators report ring-current and ring-susceptibility kernels when  $H^\alpha$ ,  $C^\alpha$ , or  $C^\beta$  are placed near a side-chain ring. Dispersion reports the short-range  $r^{-6}$  contact tensor to aromatic vertices. HydrationShell and EEQ supply solvent and charge context rather than shielding tensors.

Thus the categorical statement is conservative. The literature says  $C^\alpha$ ,  $C^\beta$ , and  $H^\alpha$  are strong reporters of backbone and secondary-structure geometry [3, 5, 6, 9]. The code proves that it emits the environment and anisotropy kernels through which that geometry can act. It does not prove, before calibration, that those kernels reproduce a pointwise  $C^\alpha$ ,  $C^\beta$ , or  $H^\alpha$  shift.

### 3.2 Carbonyl $^{13}\text{C}'$

The carbonyl carbon has a different story from  $C^\alpha$ . It sits in a planar peptide group with a strong  $C = O$  bond and partial double-bond character across  $C' - N$ . Its shielding tensor is therefore tied to the peptide plane, the carbonyl bond axis, and the hydrogen bond accepted by the oxygen. Changing  $\psi$  of this residue

and  $\phi$  of the next reorients the peptide plane the carbonyl sits in — the leading backbone-torsion effect — and hydrogen bonding to the carbonyl oxygen moves the tensor further; nearby anisotropic groups above and below the plane contribute more weakly.

Ab initio work by de Dios, Pearson, and Oldfield made this categorical separation visible: backbone torsions, hydrogen bonds, side-chain geometry, and electrostatic terms contribute differently by nucleus [3]. Oldfield’s later carbon-13 shielding surveys extend the same point across amino acid types and protein comparisons [4, 15]. Solution and solid-state CSA measurements on carbonyl, nitrogen, and amide-proton nuclei then show that these are tensor observables, not only isotropic shifts [16].

The calculator connections are direct. McConnell’s peptide  $C = O$  and peptide  $C - N$  categories place anisotropic bond kernels around the carbonyl carbon. HBond uses DSSP backbone hydrogen bonds to build a tensor along the donor  $N$  to acceptor  $O$  axis; for a carbonyl acceptor, that axis is the line along which the carbonyl electronic structure is being polarized. LarsenHBond is the local empirical exception: it looks up DFT-derived hydrogen-bond shielding terms in ppm from distance and angular coordinates, then rotates the tensor into the protein frame. APBSField and WaterField report the field-gradient terms that a Buckingham-style coefficient could multiply. The ring-current and dispersion calculators act when aromatic side chains come close enough for through-space magnetic or contact effects.

For  $C'$ , then, the categorical object is not “the carbonyl shift”. It is a set of axes: the peptide plane, the  $C = O$  bond, the  $C' - N$  bond, the hydrogen-bond line into the oxygen, and nearby aromatic or electrostatic source directions. The architecture paper says these axes are shadows of one shielding response. The categorical view says why this atom was split out in the first place.

## 4 The Amide Pair: $^1\text{H}^N$ and $^{15}\text{N}$

### 4.1 Amide proton

Backbone amide protons are the category where the hydrogen-bond term is most isolable. An amide hydrogen lies on a polar  $N - H$  bond. When it approaches an acceptor oxygen, the donor and acceptor electron densities reorganize. In shielding language, the electronic current response that screens the proton changes because the ground-state electron cloud has changed.

The historical measure exists because amide-proton shifts were too sensitive to hydrogen bonding to be explained by residue identity alone. Modern DFT and empirical models keep that term explicit. Parker, Houk, and Jensen showed that cooperative hydrogen bonding is a key determinant of backbone amide-proton shifts [32]. Moon and Case built an amide-hydrogen model that kept hydrogen-bond, ring-current, susceptibility, and electrostatic terms together [33]. ProCS15 and its lineage parameterize hydrogen bonding with DFT-derived grids indexed by local geometry [14].

The geometric variables in LarsenHBond are the same variables the chemistry talks about:

$$r = \|\mathbf{o} - \mathbf{h}\|, \quad \theta = \angle(\text{H} - \text{O} - \text{C}), \quad \rho = \text{dihedral}(\text{H} - \text{O} - \text{C} - \text{T}).$$

$\mathbf{h}$  is the donor-hydrogen position.  $\mathbf{o}$  is the acceptor oxygen.  $\text{C}$  is the heavy atom bonded to the acceptor oxygen, and  $\text{T}$  is the third atom that defines the acceptor-side plane.  $r$  is the hydrogen-to-oxygen distance.  $\theta$  says whether the hydrogen approaches along a chemically plausible acceptor direction.  $\rho$  says how the donor sits relative to the acceptor plane. LarsenHBond uses these numbers to interpolate a tensor-valued DFT table and then rotates the tensor into the protein frame.

HBond is deliberately more primitive. DSSP supplies a backbone hydrogen-bond assignment [31]. The calculator resolves it to donor  $N$  and acceptor  $O$ , places a source at the midpoint, and uses the same

McConnell-family tensor as the bond and ring-susceptibility calculators:

$$M_{ab} = \frac{9c \hat{d}_a \hat{h}_b - 3\hat{h}_a \hat{h}_b - \left(3\hat{d}_a \hat{d}_b - \delta_{ab}\right)}{r^3}.$$

$\hat{h}$  is the donor- $N$  to acceptor- $O$  unit direction.  $\hat{d}$  points from the hydrogen-bond midpoint to the target atom.  $c = \hat{d} \cdot \hat{h}$ , and  $r$  is the midpoint-to-target distance. This tensor is not a DFT hydrogen-bond shielding. It is the orientation-resolved inverse-cube proxy that can be calibrated.

Amide protons are also ring-current and electric-field reporters. A nearby Phe, Tyr, or Trp ring can shield or deshield  $H^N$  — and a His imidazole more weakly, its ring current smaller and protonation-dependent — and benchmark work has measured the magnitude of ring-current effects on amide-proton shifts [25]. APBSField and WaterField supply field and field-gradient descriptors for the same proton. HydrationShell records whether the atom is solvent-exposed and how nearby waters orient. The empirical predictors use the same mixture: hydrogen bond, ring current, electric field, and local geometry [11, 9, 12].

## 4.2 Backbone $^{15}\text{N}$

$^{15}\text{N}$  is the stubborn backbone nucleus because it belongs to the same amide group as  $H^N$  but sees more of the peptide electronic structure. Its shielding depends on the  $N-H$  bond, the  $C'-N$  peptide bond, the adjacent  $C^\alpha$ , hydrogen bonding, electrostatics, and secondary-structure geometry. It also has a large chemical-shift anisotropy, so tensor orientation matters in relaxation and aligned-media measurements.

This is why  $^{15}\text{N}$  often carries larger errors than the carbon and proton channels in predictors, even when the same structure is supplied [9, 10, 12, 14]. Dynamics can be visible in this channel: enhanced sampling improved chemical-shift prediction — particularly for  $^{15}\text{N}$ ,  $C^\alpha$ , and  $C^\beta$  — in Markwick and co-workers' study of heterogeneous protein dynamics [40]. CSA measurements and quantum studies also show that the nitrogen tensor has meaningful site-specific magnitude and orientation relative to the  $N-H$  bond [17, 18, 16].

The calculators therefore meet  $^{15}\text{N}$  as a multi-source problem. HBond and LarsenHBond encode hydrogen-bond geometry. McConnell encodes the nearby peptide  $C-N$ ,  $C=O$ , and other bond anisotropies. APBSField, WaterField, and EEQ provide electrostatic field and charge context. Ring-current calculators act when aromatic rings sit near the amide group. HydrationShell reports whether water and ions make the local dielectric environment unlike the buried peptide interior. The architecture's single kernel is still present, but the category is crowded: no single raw geometric feature should be expected to be a pointwise  $^{15}\text{N}$  shift.

## 5 Aromatic Rings and Ring-Proximal Nuclei

### 5.1 Why ring current became a measure

Aromatic shifts forced the field to name a new source. Benzene-like rings do not behave as a set of isolated local bonds in an applied magnetic field. The  $\pi$  electrons are delocalized. A field normal to the ring can induce a circulating response, and a circulating current creates a secondary magnetic field. Pople's early ring-current picture, Johnson and Bovey's two-loop model, and Haigh-Mallion surface theories are successive classical reductions of that electronic response [19, 21, 22]. Case's protein and nucleic-acid calibration made the biological connection explicit by comparing empirical ring-current formulas with DFT shielding calculations [23]. Boyd and Skrynnikov extended the discussion from isotropic ring-current shifts to shielding anisotropy [24].

The three ring-current calculators encode three versions of the same category. BiotSavart evaluates the magnetic field from Johnson–Bovey-style current loops with unit current and stores

$$G_{ab} = -10^6 B_a^{(1)} n_b.$$

$B_a^{(1)}$  is component  $a$  of the magnetic field from a unit current at the target atom.  $n_b$  is component  $b$  of the ring normal. The outer product says that only the applied-field component normal to the aromatic plane drives the current in this model.

HaighMallion integrates the dipolar kernel over the ring surface, contracts the result with the ring normal, and stores the same kind of rank-at-most-one applied-field-to-secondary-field map. RingSusceptibility collapses the ring to a point anisotropic source at the ring centre, with the ring normal replacing the bond axis in the McConnell-family tensor. PiQuadrupole treats the aromatic  $\pi$  system as an axial quadrupole source and emits a fourth-derivative electric-field-gradient kernel. Dispersion uses the ring atoms as short-range aromatic contact vertices with an  $r^{-6}$  distance law, not an  $r^{-3}$  magnetic or electrostatic field law.

## 5.2 Which atoms are ring-current categories

The obvious targets are protons near rings: side-chain aliphatic protons, methyl protons,  $H^\alpha$ , and amide protons. This is the old protein ring-current problem, and modern predictors still keep ring-current descriptors because they improve exactly those local environments [11, 9, 25, 39]. Side-chain predictor work makes the same point at wider atom coverage: methyl and other side-chain hydrogen shifts are often controlled by aromatic proximity and rotameric placement [37, 38, 39].

Aromatic CH nuclei themselves have a separate local story. Their shifts contain through-bond aromatic electronic structure, substituent effects, and ring membership. The through-space ring-current calculators intentionally reject ring atoms and atoms covalently bonded to ring atoms, because those are through-bond regimes rather than field points outside a source. That filtering is a code-level categorical boundary. The calculators are built to ask how a ring affects other nuclei nearby. They are not a complete model of the ring atoms' own aromatic chemical shifts.

McConnell's aromatic bond category is the bridge between these stories. It does not model a circulating current; it models the anisotropy of covalent bonds, including aromatic bonds, as local magnetic objects. Thus aromatic neighbour effects can enter both as a ring-current source and as a bond anisotropy source. Calibration decides whether those two shadows carry independent shielding signal or redundant geometry.

## 6 Side Chains, Methyls, Heteroatoms, and Contact Geometry

Side-chain chemical shifts are categorical because side chains move by rotamers. A  $\chi_1$  or  $\chi_2$  jump does not merely change an internal label. It moves methyl groups, aromatic rings, charged groups, donors, acceptors, and sulfur atoms through space. The shielding changes because the local electronic environment and the through-space source geometry change.

Methyl shifts show this clearly. CH3Shift models methyl shifts with rotameric, dihedral, ring-current, magnetic-anisotropy, electric-field, and distance terms [37]. PPM is a physical, chemistry-based side-chain and backbone predictor for conformational ensembles, drawing on ring-current, anisotropy, electric-field, and dihedral descriptors of the same categorical kind [38]. UCBSHift 2.0 extends this categorical view to side-chain atom types with physics-inspired features and a feature-importance analysis [39]. These are scalar predictors, but their feature categories match the physical suite.

The calculators touch side chains through explicit geometry. McConnell has side-chain  $C = O$ , aromatic, and other side-chain bond categories; disulfide and unhandled bonds enter the total tensor but not a

separate named per-category sum. Dispersion reports short-range aromatic vertex contacts:

$$K_{ab} = \frac{S(r)}{r^6} \left( 3\hat{d}_a\hat{d}_b - \delta_{ab} \right).$$

$\hat{d}$  is the unit vector from aromatic vertex to target atom.  $r$  is the vertex-to-target distance.  $S(r)$  is the code's smooth cutoff function. The matrix is symmetric and traceless; it records contact direction, while the scalar  $S(r)/r^6$  records contact strength.

Ring-current calculators act when a side-chain atom is near an aromatic plane. APBSField and WaterField act when a side-chain atom sees a strong field or field gradient from charges, solvent, or ions.

HydrationShell reports water and ion arrangement around the atom. EEQ supplies geometry-dependent partial charges and coordination numbers. LarsenHBond can include side-chain acceptor geometry when it appears in its donor/acceptor enumeration, but its output is a local ProCS-style hydrogen-bond term, not a general side-chain shift predictor.

Dispersion is the contact member of this same side-chain category. Its historical source is London's  $r^{-6}$  interaction between correlated electronic fluctuations, while modern dispersion corrections make the atom-pair coefficient chemically and environmentally dependent [34, 35, 36]. The calculator deliberately stores only the geometry: nearby aromatic vertices, contact directions, and the switched  $r^{-6}$  strength.

For sulfur, hydroxyl, carboxylate, amide side chains, and aromatic heteroatoms, the first-principles mechanism is the same but the coefficients differ. A sulfur atom is more polarizable than oxygen or nitrogen. A carboxylate carries a strong local electric field. An imidazole nitrogen changes electronic structure with protonation. The present calculators can emit the geometry around those atoms; they do not by themselves attach atom-specific response coefficients for every side-chain nucleus.

## 7 The Geometry Axis

### 7.1 Backbone $\phi, \psi$ and secondary structure

Ramachandran geometry is the strongest organizing geometry for backbone carbon and  $H^\alpha$  shifts. The archaeology runs from secondary-structure chemical shift patterns, through CSI, through Spera–Bax surfaces, through TALOS and SPARTA-style database geometry [5, 6, 7, 8, 9]. The first-principles mechanism is that  $\phi, \psi$  rotate peptide planes, bond anisotropies, hydrogen-bond opportunities, and electrostatic sources around each observed nucleus.

The calculators encode the consequences, not a discrete Ramachandran table. When  $\phi, \psi$  change, McConnell source axes move, HBond assignments and distances can change, ring positions relative to  $H^\alpha$  can change, and APBS/Water field gradients can change. Secondary structure is therefore not a single input field in the shielding-kernel suite. It is a geometry pattern that many source lists inherit.

### 7.2 Hydrogen-bond geometry

Hydrogen-bond geometry is categorical because it has a donor, an acceptor, a distance, and an angle. The physical mechanism is polarization of donor and acceptor electronic structure. The empirical state of the art keeps that geometry explicit: SHIFTX, SPARTA, SPARTA+, SHIFTX2, and ProCS15 all include hydrogen-bond descriptors or DFT hydrogen-bond terms [11, 9, 10, 12, 14].

HBond and LarsenHBond are the two code answers. HBond uses DSSP backbone partners and emits a unit-weight inverse-cube tensor on the  $N \cdots O$  axis. LarsenHBond searches donor hydrogens and acceptor oxygens directly, indexes DFT grid tensors by  $r, \theta, \rho$ , and writes tensor terms in ppm. The first is a geometric kernel awaiting calibration. The second is a local empirical shielding contribution, still incomplete as a total shift.

### 7.3 Aromatic-ring geometry

Aromatic-ring geometry has three coordinates that matter immediately: distance from the ring centre or vertices, height above the ring plane, and azimuth relative to the ring atoms. BiotSavart, HaighMallion, and RingSusceptibility all need the ring normal. PiQuadrupole needs the same normal and the height  $h = d \cdot \hat{n}$ . Dispersion needs the individual ring-vertex directions. These are not interchangeable approximations; they are different geometric readings of the same aromatic category.

The literature history explains why several readings survived. Point susceptibility is cheap and captures the far-field two-lobed pattern. Johnson–Bovey loops preserve the current-loop picture. Haigh–Mallion integrals preserve a surface response. Case-style calibration asks which approximation carries the measured shielding signal for biological aromatic rings [23]. The code keeps these alternatives separate so DFT calibration can test them rather than choosing by tradition.

### 7.4 Electric-field and solvent geometry

Electric-field geometry is the field at the nucleus and the curvature of that field around the nucleus. Buckingham’s expansion can be summarized as

$$\Delta\sigma_I \approx A_I E_{\parallel} + \gamma_I V_{ab} Q_{ab}^{(I)} + \dots$$

$E_{\parallel}$  is the electric-field component along a nucleus-specific local axis.  $V_{ab}$  is an electric-field-gradient tensor.  $Q_{ab}^{(I)}$  denotes the local tensor direction to which nucleus  $I$ ’s shielding response is sensitive.  $A_I$  and  $\gamma_I$  are response coefficients. The code emits the field and the field-gradient geometry; it does not know  $A_I$  or  $\gamma_I$  until calibration.

APBSField obtains  $E$  and the field-gradient tensor by finite-differencing a Poisson–Boltzmann potential grid [30]. WaterField obtains the same kind of tensor by analytic point-charge sums over explicit water sites. EEQ assigns partial charges that can feed this kind of field construction. HydrationShell records first-shell water and ion geometry without pretending to be a tensor. The protein-shift literature uses the same route when it compares chemical-shift perturbations with electric fields or dielectric models [27, 28, 29].

### 7.5 Side-chain $\chi$ angles

Side-chain  $\chi$  angles are the geometry of rotameric exposure. A rotamer can place a methyl group under a ring, swing an acceptor into a hydrogen bond, move a charged group across a backbone amide, or turn a sulfur contact toward a target nucleus. The first-principles mechanism is not one new force. It is the same response terms moved by a side-chain torsion.

The code sees  $\chi$  through Cartesian consequences: ring-current distance, bond-anisotropy axes, electric fields, water exposure, and dispersion contact. The scalar literature sees the same thing as feature importance and fitted rotamer terms in CH3Shift, PPM, SHIFTX2, and UCBSHIFT 2.0 [12, 37, 38, 39]. This is why side-chain prediction is not a minor extension of backbone prediction. It multiplies the number of atom categories and the number of geometry stories.

## 8 What This Categorical Grounding Proves

The categorical literature proves that protein chemical shifts are geometry signals, but not one geometry signal.  $C^\alpha$ ,  $C^\beta$ ,  $H^\alpha$ ,  $C'$ ,  $H^N$ ,  $^{15}\text{N}$ , aromatic protons, methyl protons, side-chain nitrogens, and heteroatoms do not share the same dominant determinants. That is why empirical machines are categorical.

The physics architecture proves that many of those categories can be written as classical shadows of one response problem. The recurrent kernel  $D_{ab} = \partial_a \partial_b (1/\rho)$  appears as bond anisotropy, ring susceptibility, hydrogen-bond proxy, electric-field gradient, and the angular part of dispersion contact. Ring-current line and surface integrals, ProCS hydrogen-bond grids, charge-equilibration, and hydration descriptors sit around that same response problem.

The code proves a narrower statement. For each conformation, it emits traceable geometric objects with known units, source lists, distance laws, near-field exclusions, and tensor packing. Those objects can be stratified by atom type and geometry category. Calibration can then ask whether the object correlates with DFT shielding changes or experimental chemical shifts. A raw kernel value is not a shielding value. A high correlation is not a pointwise identity. The value of the categorical grounding is that it tells us what kind of signal each nucleus could reasonably carry before the regression begins.

## 9 Reference Status

Modern support used as evidence for protein shielding determinants, geometry-to-shift prediction, hydrogen bonding, ring-current effects, electric-field effects, side-chain prediction, and dynamics was checked against Crossref DOI metadata and Europe PMC abstract records where available. This abstract-confirmed group includes de Dios 1993, Oldfield 1995, Williamson 1990, Wishart chemical-shift index 1992, TALOS 1999, SPARTA 2007, SPARTA+ 2010, SHIFTX 2003, SHIFTX2 2011, CamShift 2009, ProCS15 2015, Sun, Sanders, and Oldfield 2002, Yao 2010, Loth 2005, Brender 2001, Parker 2006, Moon and Case 2007, Case 1995, Boyd and Skrynnikov 2002, Christensen 2011, Hass 2008, Kukic 2013, Avbelj 2004, Sahakyan 2011, PPM 2012, UCBSHIFT 2.0 2024, Markwick 2010, Baker 2001, Grimme 2010, and Wagner and Schreiner 2015.

Facelli 2011 is reused as the tensor-theory review from `physics-architecture.tex`. Its DOI metadata and open full text are available, but the Europe PMC record did not expose an `abstractText` field during this pass.

Spera and Bax 1991 is DOI-confirmed through Crossref but Europe PMC returned no abstract record during this pass. It is cited as a central DOI-confirmed determinant paper because the  $C^\alpha/C^\beta \phi, \psi$  surface is part of the field's categorical history and is also carried through the abstract-confirmed SPARTA discussion.

The pre-digital and old-canon mechanism papers are treated as named lineage, as in `physics-architecture.tex`: Ramsey 1950, Pople 1956, McConnell 1957, Johnson and Bovey 1958, Haigh and Mallion 1979, Buckingham 1960, London 1937, and Kabsch and Sander 1983 have DOI metadata support, but a modern Europe PMC abstract was absent or incomplete in this pass. They are anchored here by the modern reviews, calibrations, and implementations cited around them.

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